

Theoretical Computer Science Toolkit

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1 Preliminaries

1.1 Big O and friends

Recall that $f(x) = O(g(x))$ (as $x \rightarrow \infty$) $\iff \exists c, x_0$ s.t. $|f(x)| \leq c \cdot g(x)$, $\forall x \geq x_0$. **Example:** $\sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2} = O(n^2) = \frac{n^2}{2} + O(n)$.

Definition 1.1 (Harmonic Number). Let $n \in \mathbf{N}$, then we define $H_n = 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n$.

We can get an **upper bound** and a **lower bound** for harmonic number in the following way:

$$\begin{aligned} H_n &= 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n \leq \\ 1 + 1/2 + 1/2 + 1/4 + 1/4 + 1/4 + 1/4 + 1/8 + \dots &\leq \\ \lfloor \log_2 n \rfloor + 1. \end{aligned}$$

$$\begin{aligned} H_n &= 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n \geq \\ 1 + 1/2 + 1/4 + 1/4 + 1/8 + 1/8 + 1/8 + 1/8 + 1/16 \dots &\geq \\ 1/2 \lceil \log_2 n \rceil + 1. \end{aligned}$$

Since that $H_n \leq O(\log n)$ and $H_n \geq \Omega(\log n)$, we can conclude that $H_n = \Theta(\log n)$.

Definition 1.2 ($f(x) \sim g(x)$). We say that $f(x) \sim g(x)$ as $x \rightarrow \infty$ if $\frac{f(x)}{g(x)} \rightarrow 1$.

We recall the definition of a Taylor Series:

Theorem 1.1 (Taylor Series). If $f(x)$ is analytic at $x = a$, then it can be represented by its Taylor series:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n,$$

which converges to $f(x)$ for all x within the radius of convergence.

Definition 1.3 (e^x). We define the function $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$, $\forall x \in \mathbf{R}$.

For example, the function $\ln 1 + x = \sum_{n=0}^{\infty} (-1)^{n-1} \frac{x^n}{n!}$, for $|x| \leq 1$. Additionally, we can use this equivalence to approximate $\ln 1 + x = \sum_{n=0}^{\infty} (-1)^{n-1} \frac{x^n}{n!} \sim x$ when $x \rightarrow 0^+$. We can also exponent this and get: $\exp x \sim 1 + x$.

Definition 1.4 (Central Binomial Coefficient). We define the central binomial coefficient as $C_n := \binom{n}{n/2}$. We notice that the probability of getting half of n flipped coins is $\frac{C_n}{2^n}$.

Proposition 1.2. $C_n = \Theta(\frac{2^n}{\sqrt{n}})$.

Proof. We can bound above and below the expression. Lets do a simple example to see this. $C_{10} = \frac{10 \cdot 9 \cdot 8 \cdot \dots \cdot 2 \cdot 1}{5 \cdot 4 \cdot \dots \cdot 1 \cdot 5 \cdot 4 \cdot \dots \cdot 1}$. Then we divide by 2^n and get: $C_{10}/2^n = \frac{10 \cdot 9 \cdot 8 \cdot \dots \cdot 2 \cdot 1}{10 \cdot 8 \cdot \dots \cdot 2 \cdot 10 \cdot 8 \cdot \dots \cdot 2}$. Furthermore we can potentiate to $(\cdot)^2$ and get: $(\frac{C_n}{2^n})^2 = \frac{10 \cdot 10 \cdot 9 \cdot 9 \cdot 8 \cdot 8 \cdot \dots \cdot 2 \cdot 2 \cdot 1}{10 \cdot 10 \cdot 10 \cdot 10 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdot \dots \cdot 2 \cdot 2 \cdot 2 \cdot 2} = \frac{9 \cdot 9 \cdot 7 \cdot 7 \cdot 5 \cdot 5 \cdot 3 \cdot 3}{10 \cdot 10 \cdot 8 \cdot 8 \cdot 6 \cdot 6 \cdot 4 \cdot 4 \cdot 2 \cdot 2}$. From this we can get an upper bound: $(\frac{C_n}{2^n})^2 \leq \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} = \frac{1}{11} = \frac{1}{n+1}$. Also we can get a lower bound: $(\frac{C_n}{2^n})^2 \geq \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{1}{10} = \frac{1}{n}$. $\square$

1.2 Sum of Random Variables, Gaussian and Central Limit Theorem

We work with the following setup: let $X_1, X_2, \dots, X_n$ be i.i.d random variables with $\mathbf{P}[X_i = 1] = p$ and $\mathbf{P}[X_i = 0] = 1 - p = q$. Let $S_n = X_1 + X_2 + \dots + X_n$, then $\mathbf{E}(S_n) = \mathbf{E}(X_1) + \dots + \mathbf{E}(X_n) = pn$.

2 Boolean Functions

A boolean function is perhaps the most simplest function in computer science.

Definition 2.1 (Boolean Function). A function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is called a boolean function.

There are different ways to measure the complexity of a boolean function. Here we present four: **decision tree complexity, sensitivity, certificate complexity and degree.**

Definition 2.2 (Decision Tree complexity). The decision tree complexity $d(f)$ of a boolean function is the depth of the smallest tree that computes a boolean function.

Definition 2.3 (Sensitivity). The hypercube is a graph that connects inputs of a boolean function f which differ by 1 bit. The nodes (inputs) in the hypercube are colored with the output of the boolean function. The sensitivity complexity $s(f)$ is defined as the maximum number of neighbours with different color in the hypercube.

Definition 2.4 (Certificate Complexity). A certificate is a set of coordinates in the input that once we fixed those coordinates, the value of the function is determined. The certificate complexity $c(f)$ is the smallest number such that every input x has a certificate of size $c(f)$.

Definition 2.5 (Degree Complexity). Any function $f : \{0, 1\}^n \rightarrow \mathbb{R}$ can be expressed as a multilinear polynomial. The degree complexity $deg(f)$ of a boolean function is the degree of this polynomial.

We can establish bound between these complexity measures:

Proposition 2.1. The following bounds hold:

- $s(f) \leq c(f)$
- $c(f) \leq d(f)$
- $deg(f) \leq d(f)$
- $d(f) \leq c(f)^2$

3 Fourier Analysis of Boolean Functions

Let start motivating the topic constructing a polynomial that represents a boolean function.

Example 3.1 (Fourier representation of Max boolean function). Let $Max_2 : \{-1, 1\}^2 \rightarrow \{-1, 1\}$, which takes the max of the input. We construct the fourier representation in the following way: $Max_2(x_1, x_2) = \frac{1}{2} + \frac{1}{2}x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_1x_2$.

Theorem 3.1. Any function $f : \{+1, -1\}^n \rightarrow \mathbb{R}$ is uniquely expressible as a multilinear polynomial.

We represent function as $f(x) = f(x_1, \dots, x_n) = \sum_{S \subseteq [n]} \hat{f}(S) \prod_{i \in S} x_i$. In this way we can see that the fourier expansion is a linear combinations "XOR"s functions. This is because $\prod_{i \in S} x_i$ is the parity or XOR of bits in S (considering that x_i are -1 and 1s).

Definition 3.1. Let $S \subset [n]$, we define $X_S : \{0, 1\} \rightarrow \{0, 1\}$ as $X_S(x) = \prod_{i \in S} x_i = x^S$.

Let V the vector space of functions $f : \{-1, +1\} \rightarrow \mathbb{R}$. We notice that the dimension of this vector space is 2^n .

4 Circuits

Definition 4.1 (Circuit). A circuit is a Directed Acyclic Graphs (DAGs) that computes Boolean Functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$. The nodes of a circuit are called gates. A circuit can be seen as a tree, where the leafs are input gates that receives an input $x = (x_1, x_2, \dots, x_n)$ and the root generates an output bit.

Definition 4.2 (Size and Depth of Circuit). The Size and Depth of a circuit C are defined as the number of non-input gates and the longest input-output path, respectively.

Example 4.1. Some common basis are the following: $\mathcal{B}_1 = \{\neg, \text{binary } \vee, \text{binary } \wedge\}$, $\mathcal{B}_2 = \{\text{all } g : \{0, 1\}^2 \rightarrow \{0, 1\}\}$, $\mathcal{B}_3 = \{\neg, \text{all fan-in } \vee, \text{all fan-in } \wedge\}$. Notice that $\mathcal{B}_3$ allow for constant depth computations. For example, we can add two n -bits numbers in constant depth. Instead for $\mathcal{B}_1$, we need at least a depth of $\log n$ to look all bits.

Example 4.2. We study the function **AND**: $\{0, 1\}^n \rightarrow \{0, 1\}$. Notice that with $\mathcal{B}_3$ we can generate a circuit with size 1. Instead with $\mathcal{B}_2$ we can generate a circuit with size $O(n)$ and depth $O(\log n)$.

Example 4.3. We study the **Palindrome** function $p : \{0, 1\}^{2^n} \rightarrow \{0, 1\}$. With $\mathcal{B}_3$ we can generate a circuit with depth 2, by generating an "equal" gate.

Definition 4.3 (Basis). We call $\mathcal{B}(C)$ the basis of a circuit C , which is a set of allowed operations in gates of a circuit C .

We have talk about constant size inputs of size n , this mean that for each input size we need to generate a different circuit. In order to accept a certain language L , we define a **circuit family** $(C_n)_{n=1}^\infty$, where the circuit C_n should accepts all words in $L \cap \{0, 1\}^n$. Now, we define some families of languages that are categorized by the size of circuit families (with $\mathcal{B}_3$) that accept them.

Definition 4.4 (Family AC^0). The collection of languages decided by a circuit family of depth $O(1)$ and size $\text{poly}(n)$.

Definition 4.5 (Family NC). The collection of languages decided by a circuit family of depth $\text{polylog}(n)$ and size $\text{poly}(n)$.

There are another notations about language decidable by circuits.

Definition 4.6 (P-Poly). The family of languages decidable by $\text{poly}(n)$ -size circuits size.

A cool proposition about circuits.

Proposition 4.1 (Has 86). There exists a language $L \in P$ with no poly -size, constant-depth circuits.

Proposition 4.2 (CLIQUE). CLIQUE requires exponential size AND/OR circuits.

5 Communication Complexity

We study the case of 2-party communication between Alice and Bob. The idea is that A and B want to compute the function $f : X \times Y \rightarrow Z$. The goal is given $x \in X$ (Alice) and $y \in Y$ (Bob) that both Alice and Bob computes $f(x, y)$. We only care about how many communication steps Alice and Bob perform. We consider that each time Alice and Bob communicate to each other a cost of 1-bit.

Example 5.1 (EQUALITY). Alice and Bob have both words from $\{0, 1\}^n$ and they try to know if these are equal. So the equality can be represented by $EQ : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$.

Proposition 5.1. $n + 1$ bits are sufficient and necessary to solve EQ.

Proof. There is an easy to detect whether $X = Y$, Alice can send the n bits from her word to Bob, then Bob knows the answer and he send it to Alice. $\square$

Example 5.2 (PARITY). Alice and Bob have both words from $\{0, 1\}^n$ and they try to compute PARITY function $PARITY(x, y) = (\sum x_i + \sum y_i) \bmod 2$.

Proposition 5.2. The communication complexity of PARITY $cc(PARITY)$ holds: $cc(PARITY) \leq 2$.

There are cases when there is a one-way communication and other cases when there is a need to turn the communication. The following is a **key example in communication complexity**:

Example 5.3 (DISJOINTNESS). Let $x, y \subseteq [n]$ the DISJOINTNESS function $DISJ : x \times y \rightarrow \{0, 1\}$, defined by $DISJ(x, y) = 1 \iff x \cap y = \emptyset$. In CS term, we can think that $f(x, y) = \neg \vee (x_1 \wedge y_1, \dots, x_n \wedge y_n)$

Proposition 5.3 (DISJOINTNESS complexity). For a deterministic and randomized communication DISJ has the following complexity: $n + 1$ and $\Omega(n)$.